Module 04 Lab - Exploratory Data Analysis**Objective:** To learn how to explore a dataset to find patterns, anomalies, and insights before modeling. EDA is like being a detective; you are looking for clues in the data that will help you build a better model.**This lab is fully coded.** Your task is to run each cell, read the detailed explanations, understand the purpose of each visualization, and then complete the experimentation section.

Part 1: Setup and Data Loading**What is Exploratory Data Analysis (EDA)?**EDA is the process of using summary statistics and visualizations to understand a dataset's main characteristics. Before you can build a model, you need to understand your data. What stories does it tell? Are there errors or missing values? Are there strong relationships between variables? EDA helps answer these questions.We will use the famous Titanic dataset for this lab. It contains information about passengers and, crucially, whether they survived the disaster.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns # Seaborn is a library built on top of Matplotlib that makes creating beautiful plots ea

# Load the dataset directly from a URL
df = pd.read_csv('https://raw.githubusercontent.com/datasciencedojo/datasets/master/titanic.csv')

print("--- First 5 Rows ---")
print(df.head())

print("--- Basic Info ---")
# .info() is a great first command. It tells us the column names, how many non-null values are in each column,
# Notice that 'Age' and 'Cabin' have missing values!
df.info()
```

```
--- First 5 Rows ---
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
--- Basic Info ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    object
 4   Sex          891 non-null    object
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
```

```
 8   Ticket       891 non-null    object
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object
 11  Embarked     889 non-null    object
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

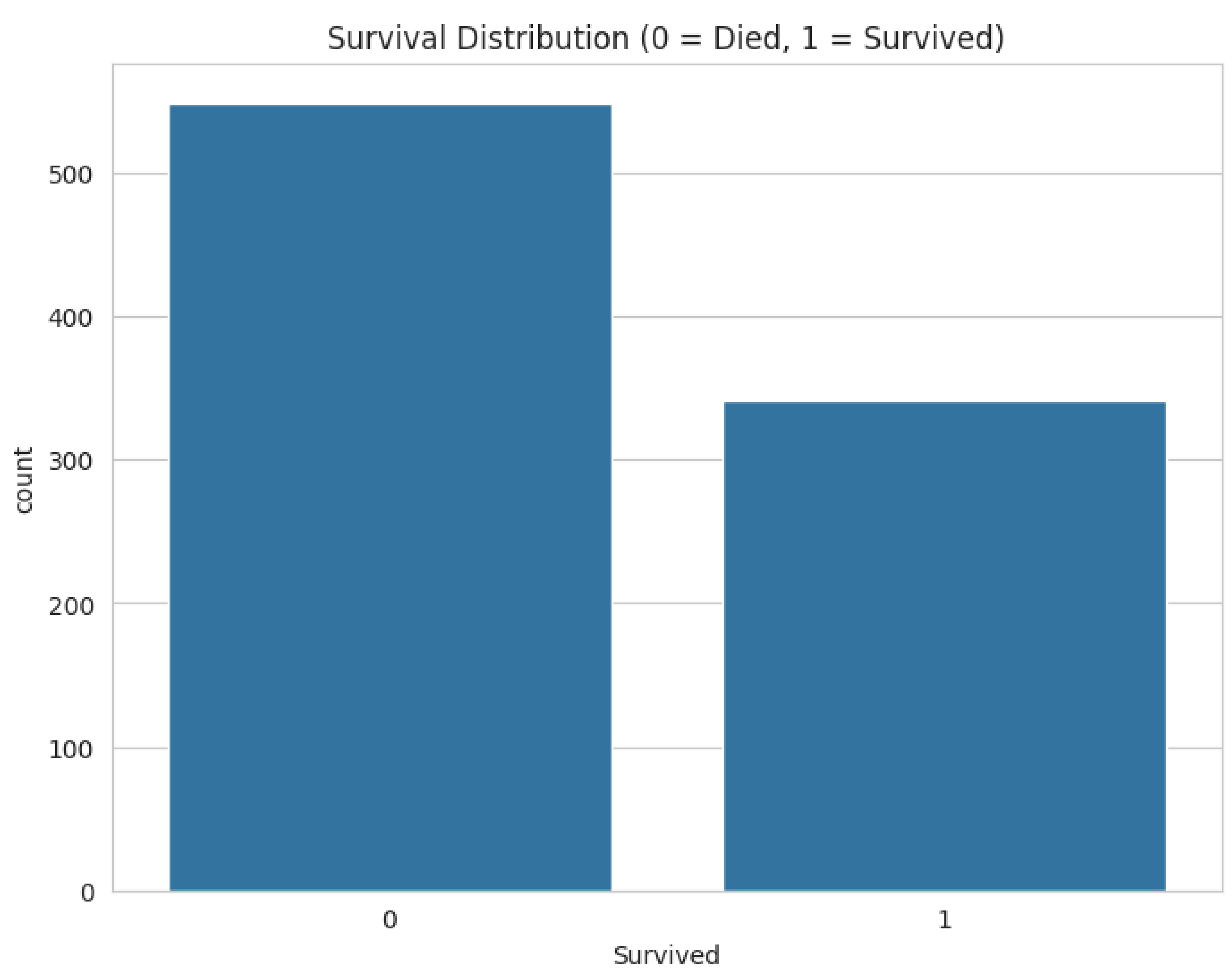
Part 2: Descriptive StatisticsLet's start by getting a high-level numerical summary of the data. The `.describe()` method is perfect for this. It calculates statistics like mean, standard deviation, min, and max for the numerical columns.

```
# Get summary statistics for numerical columnsprint("--- Descriptive Statistics ---")print(df.describe())print
```

Part 3: Visual EDA - Telling Stories with PlotsNumbers are great, but plots make patterns and relationships immediately obvious. The goal of visual EDA is to turn data into insights.**A Note on Plotting Libraries:*** **Matplotlib:** The foundational library, gives you full control over everything.* **Seaborn:** Built on Matplotlib, it provides a simpler, high-level interface for creating common statistical plots. We will use Seaborn for its ease of use and attractive defaults.

Visualization 1: How many survived?A simple `countplot` is the best way to see the distribution of a categorical variable, like our target `Survived`.

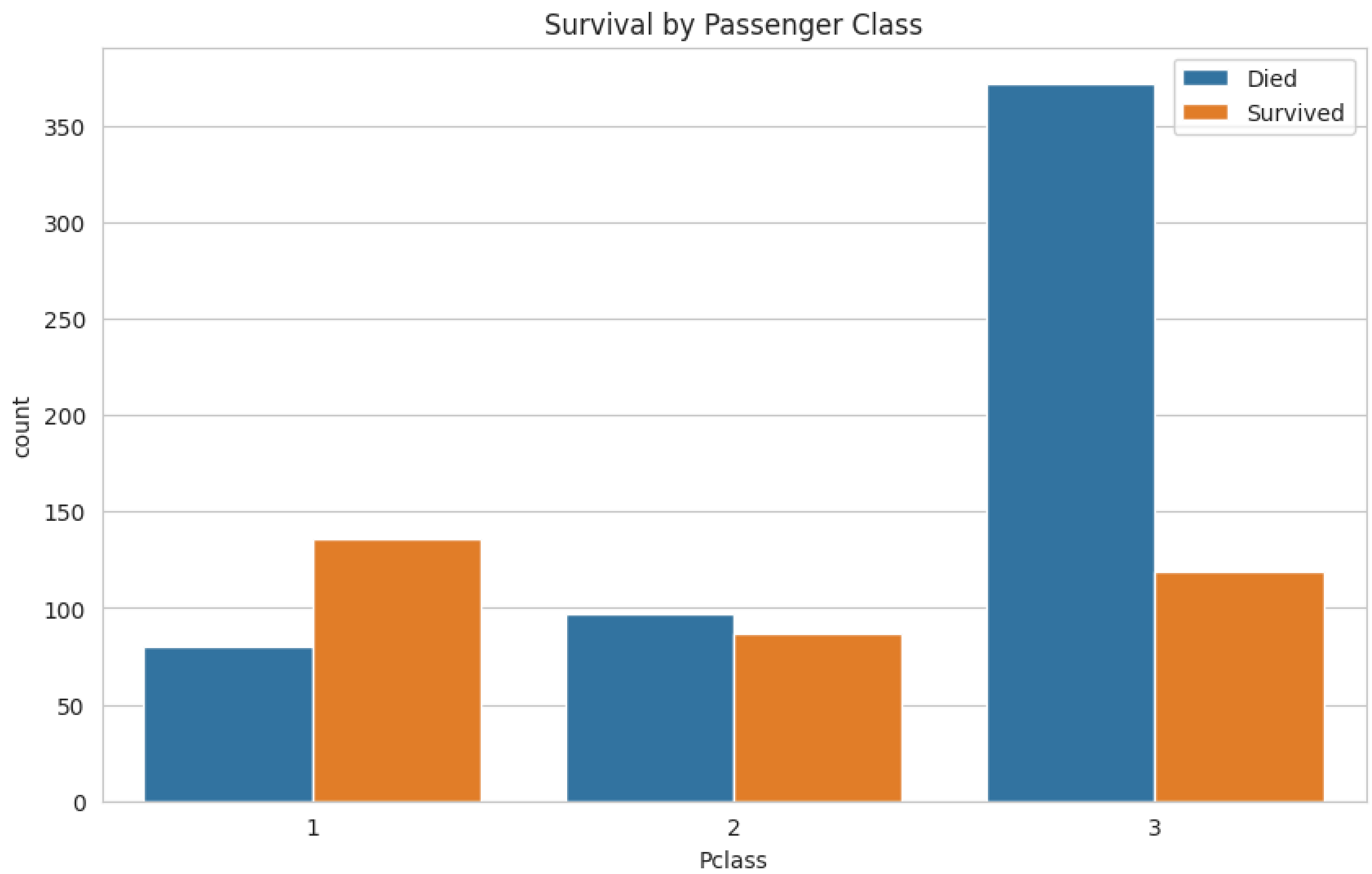
```
sns.set_style('whitegrid') # Sets a nice visual style for our plots
plt.figure(figsize=(8, 6))
sns.countplot(x='Survived', data=df)
plt.title('Survival Distribution (0 = Died, 1 = Survived)')
plt.show()
print("Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can som
```




```
Insight: Far more people died than survived. This is an example of an imbalanced dataset, which can sometimes b
```

Visualization 2: Does passenger class matter for survival?Now we want to see if there's a relationship between two variables. We can use the `hue` parameter in `countplot` to split the bars by another category.

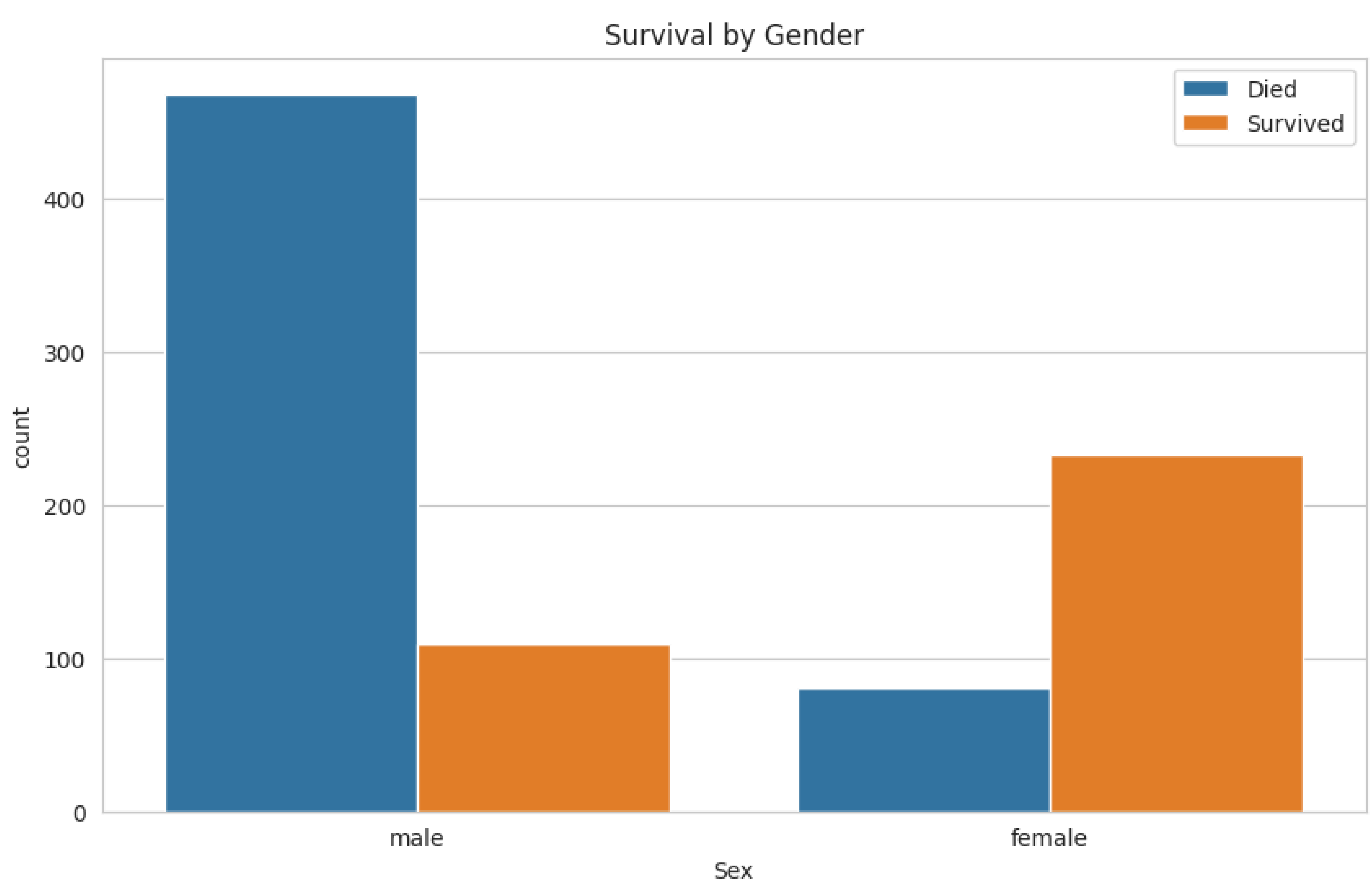
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Pclass', hue='Survived', data=df)
plt.title('Survival by Passenger Class')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compa
```




```
Insight: This is a very strong pattern. 1st class passengers had a much higher chance of survival compared to 3
```

Visualization 3: What about gender?The 'women and children first' mantra is famous. Let's see if the data supports it.

```
plt.figure(figsize=(10, 6))
sns.countplot(x='Sex', hue='Survived', data=df)
plt.title('Survival by Gender')
plt.legend(['Died', 'Survived'])
plt.show()
print("Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. Thi
```




```
Insight: The pattern is undeniable. A much higher proportion of females survived compared to males. This is ano
```

Visualization 4: How does age play a role?For a continuous variable like `Age`, a histogram is a great way to see its distribution.

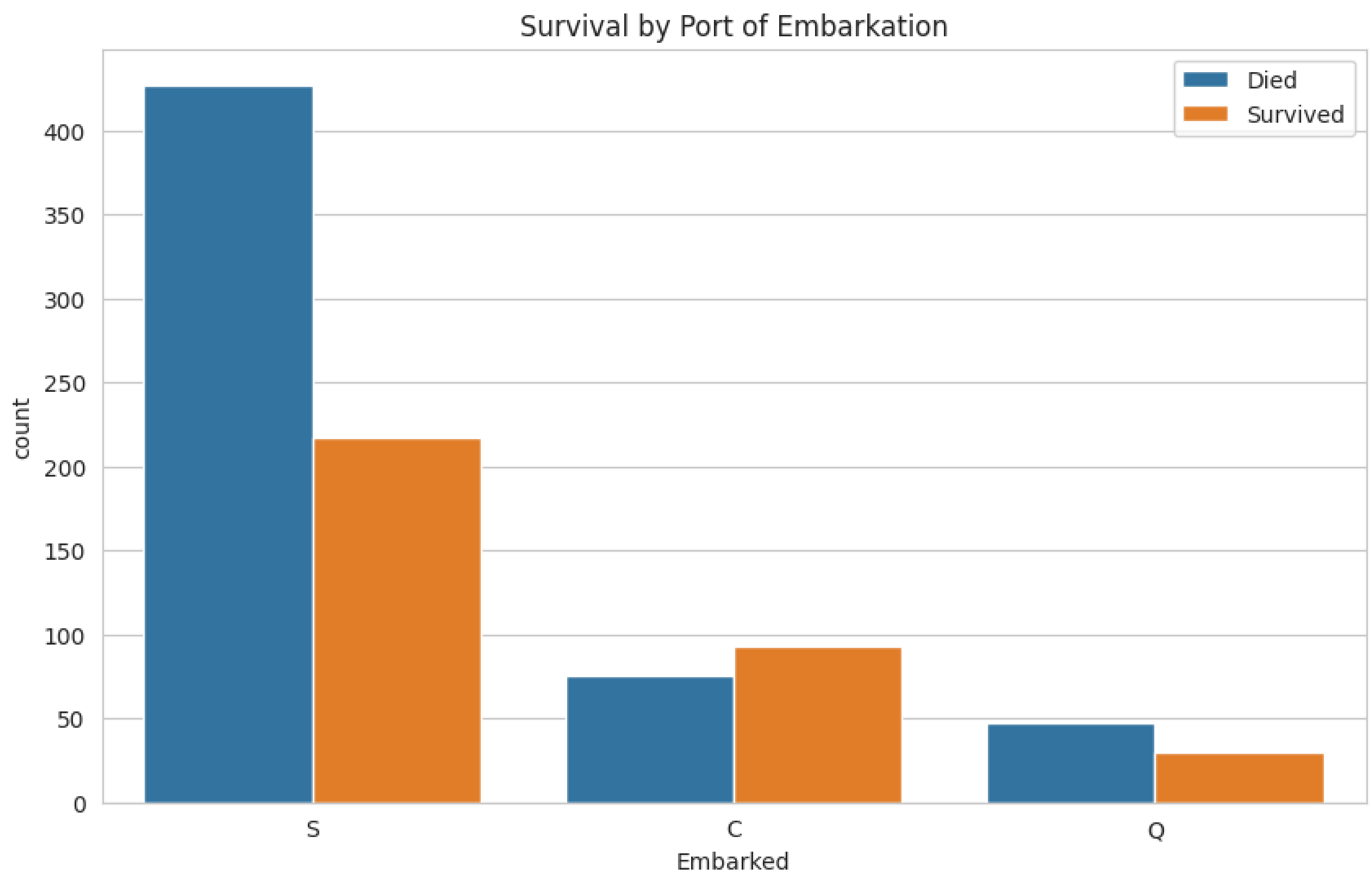
```
# A FacetGrid allows us to create multiple plots side-by-side to compare distributions.# Here, we create one h
```

Part 4: Student Experimentation**Instructions:** Create your own visualizations to explore other relationships in the data.

Experiment 1: Port of Embarkation1. The `Embarked` column tells you where a passenger boarded the ship (C = Cherbourg, Q = Queenstown, S = Southampton).2. Create a `countplot` to see how survival rates differed by the port of embarkation. Does where they boarded seem to be related to their survival?

yes, Passengers who boarded at Cherbourg (C) had a higher proportion of survivors relative to those who died compared to the other ports. In contrast, those who boarded at Southampton (S) and Queenstown (Q) saw significantly more fatalities than survivors, with Southampton having the highest absolute number of deaths due to the large majority of passengers embarking there.

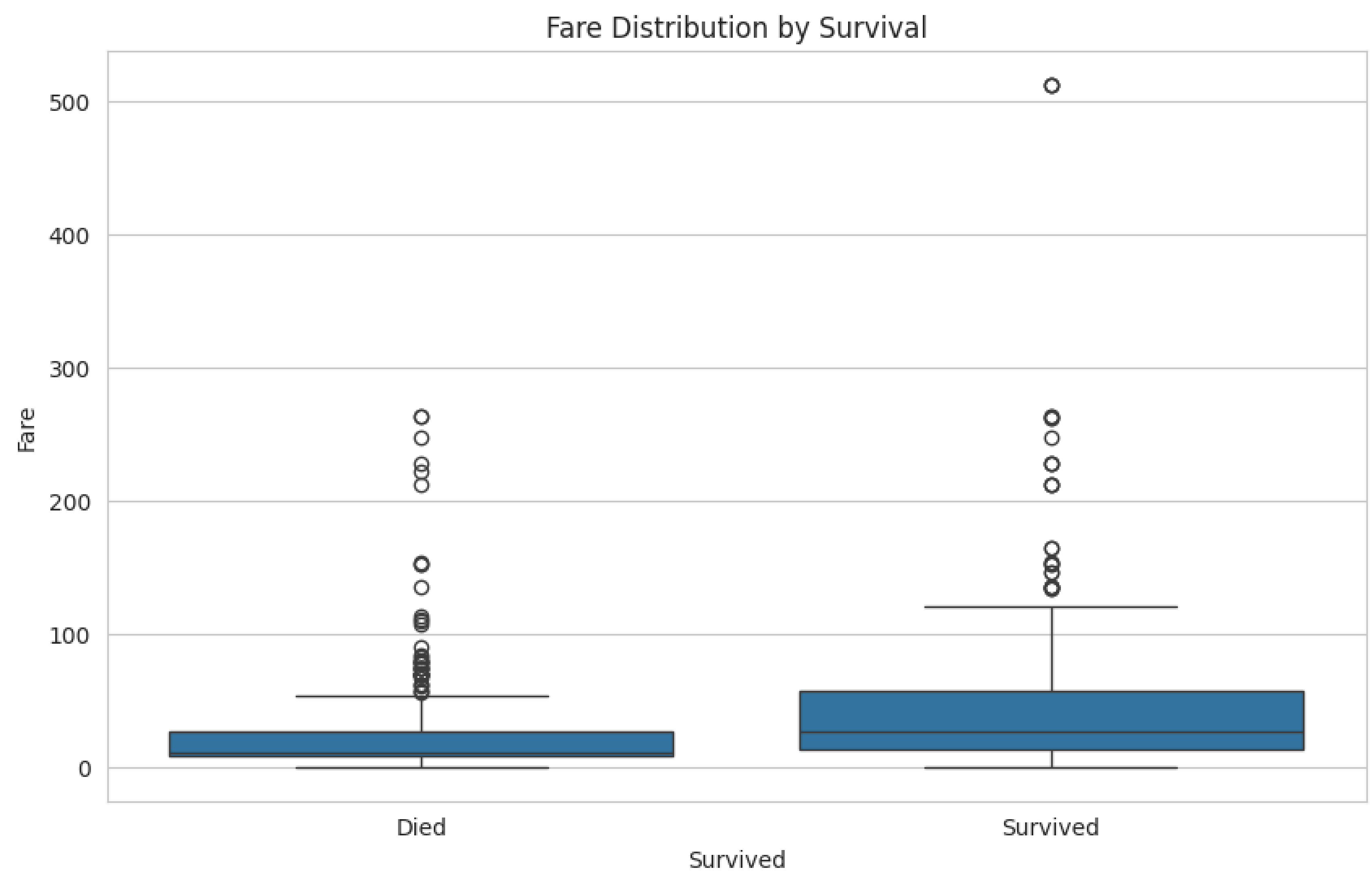
```
plt.figure(figsize=(10, 6))
sns.countplot(x='Embarked', hue='Survived', data=df)
plt.title('Survival by Port of Embarkation')
plt.legend(['Died', 'Survived'])
plt.show()
```



Experiment 2: Fare vs. Survival1. `Fare` is a continuous numerical variable.2. A `boxplot` or `violinplot` is a great way to see the distribution of fares for those who survived vs. those who didn't.3. Create one of these plots to compare the `Fare` distribution by `Survived`. What does it tell you? The box plot shows that passengers who survived generally paid significantly higher fares than those who did not survive. The median fare and the entire distribution for survivors (1) are shifted upward, and the upper outliers indicate that individuals who paid the highest ticket prices (likely first-class passengers) had a much higher likelihood of survival.

```
plt.figure(figsize=(10, 6))
sns.boxplot(x='Survived', y='Fare', data=df)
plt.title('Fare Distribution by Survival')
```

```
plt.xticks([0, 1], ['Died', 'Survived'])
plt.show()
```



📝 Knowledge Check**Instructions:** Answer the following questions in this markdown cell.1. **What is the primary goal of Exploratory Data Analysis (EDA)?** The goal is to discover the main characteristics of a data set, patterns, anomalies, missing values, and relationships between variables prior to developing a machine learning model.**2.Based on the plots in this lab, what kind of person had the best chance of surviving the Titanic?** (Describe them in terms of class, gender, and age).According to the visualizations provided by the lab, one who had the best chance of surviving was a woman in the first class or a kid. 3. **Why is it important to visualize data instead of just looking at summary statistics? What can a plot show you that a number like 'mean' or 'count' can't?** Visual representations expose the true nature of the distribution, shape, patterns, and outliers that summary statistics such as the mean or count cannot possibly explain. In other words, the mean is easily skewed by a few outliers, but histograms, box plots, and facet grids can show you multimodal distributions, peaks, and groups in your data.